

CS & IT ENGINEERING



Digital Logic

Sequential Circuits

Lecture No. 5



By- Aditya sir

Recap of Previous Lecture



Topic



→ JK FF
→ Race Around Problem.

Topics to be Covered



Topic



- 1> Master-Slave FF
- 2> D FF
- 3> T FF
- 4> FF Design



About Aditya Jain sir



1. Appeared for GATE during BTech and secured AIR 60 in GATE in very first attempt - City topper
2. Represented college as the first Google DSC Ambassador.
3. The only student from the batch to secure an internship at Amazon. (9+ CGPA)
4. Had offer from IIT Bombay and IISc Bangalore to join the Masters program
5. Joined IIT Bombay for my 2 year Masters program, specialization in Data Science
6. Published multiple research papers in well known conferences along with the team
7. Received the prestigious excellence in Research award from IIT Bombay for my Masters thesis
8. Completed my Masters with an overall GPA of 9.36/10
9. Joined Dream11 as a Data Scientist
10. Have mentored 12,000+ students & working professions in field of Data Science and Analytics
11. Have been mentoring & teaching GATE aspirants to secure a great rank in limited time
12. Have got around 27.5K followers on LinkedIn where I share my insights and guide students and professionals.





Telegram Link for Aditya Jain sir: https://t.me/AdityaSir_PW

#Q. Which of the following will be correct for the given sequential circuit?

H.W

$$\underline{Q_{t+1} = F(X, Y, Q_t)}$$

X	Y	Q(t)	Q(t + 1)
0	x	0	0
0	1	0	1
1	0	1	0
x	1	1	1

A $X Q(t)' + Y' Q(t)$

B $X + Y' Q(t)$

C $X Q(t) + Y' Q(t)'$

D $X + Y' Q(t)'$



Topic : JK Flip Flop



NOTE: Level triggered JK FF suffers from the problem of RACE AROUND.



Topic : RACE AROUND Problem

When $J = K = 1$ is applied to the level sensitive of JK FF, then within the duration of pulse o/p of the FF Toggle more than one times are called RACE AROUND problem.



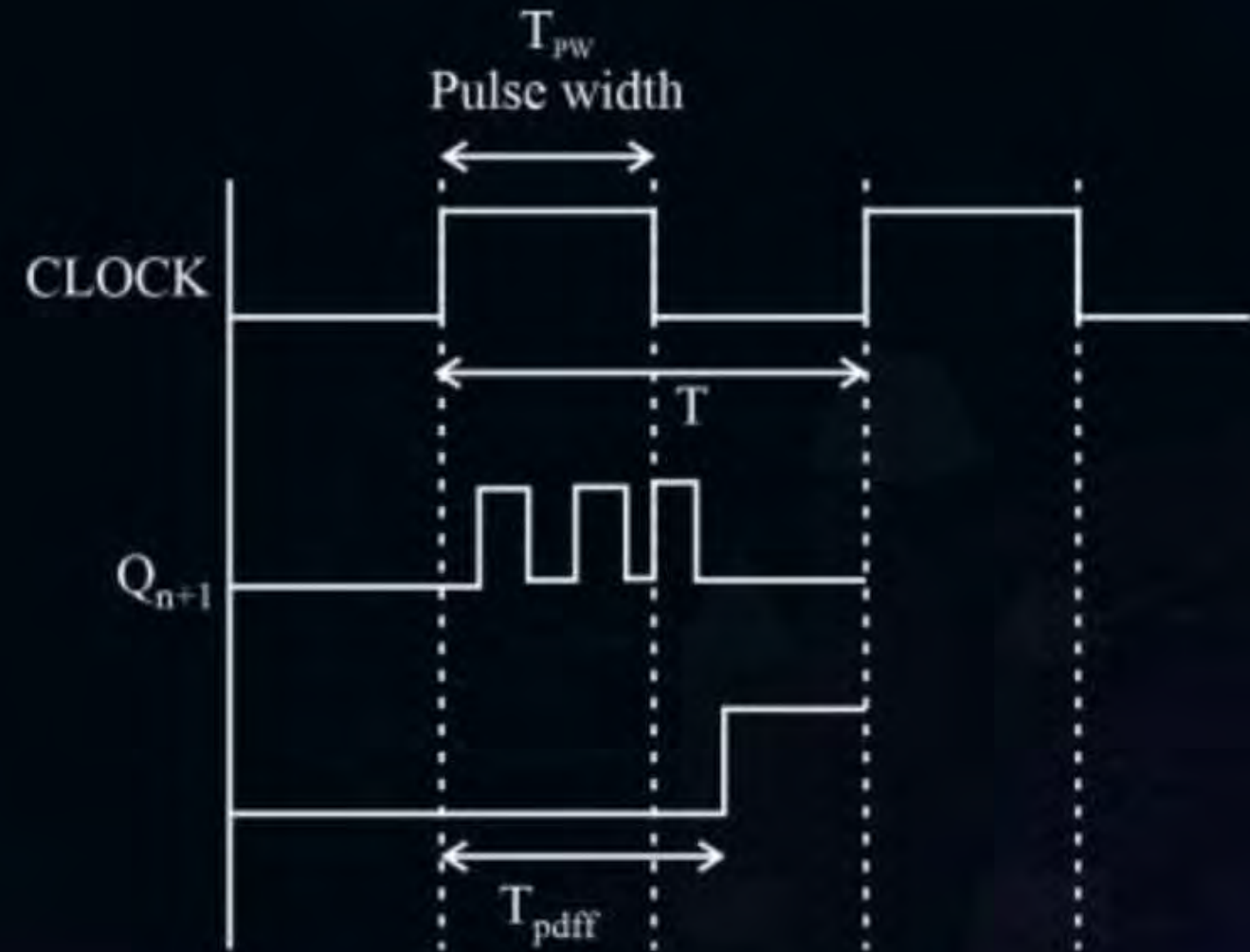
Topic : RACE AROUND Problem

To avoid the RACE AROUND problem



Solutions to Race Around:

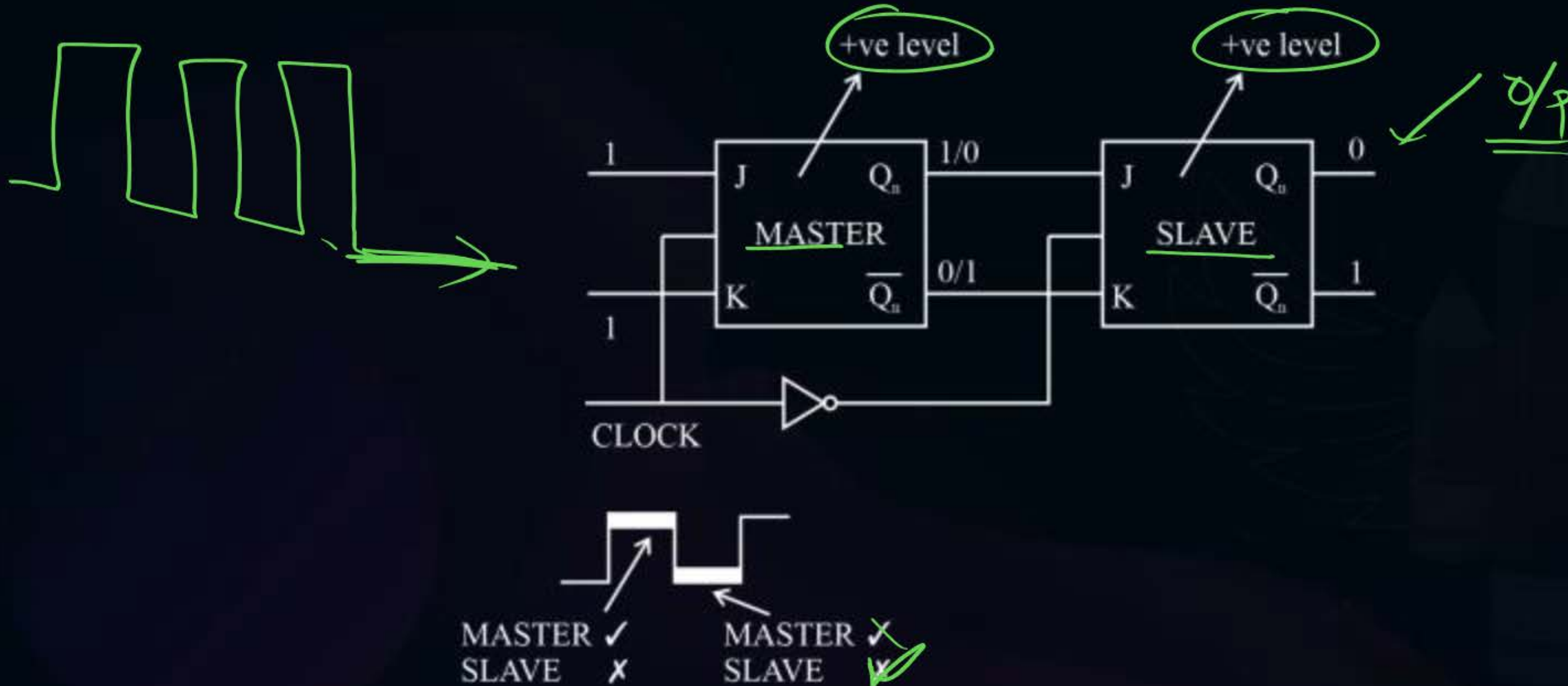
- (1) $T_{pw} < T_{pdff} < T_{CLK}$
- (2) By Master slave FF.





Topic : Master Slave FF

- (1) Inverted clock is applied to the slave as compared to master.
- (2) M-S FF is used to store single bit because o/p is taken only from slave.





Topic : Master-Slave FF



- i. In master-slave FF inverted CLK is given to master and slave.
- ★ ii. Master-slave FF is used to store single bit because output is taken only from the slave.
- iii. By the operation it seems that master is level triggered whereas slave is negative edged triggered.

1) S-R FF (Set-Reset FF)

2) JK FF

3) D FF (Delay FF)

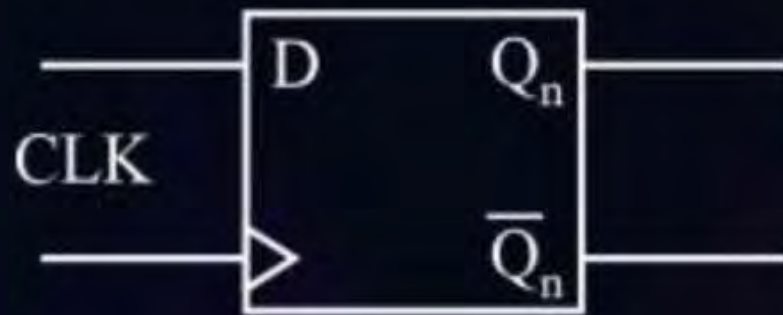
4) T FF (Toggle)



Topic : D Flip Flop

- (i) It is known as Delay FF or Transparent FF.
- (ii) In the D FF whatever the input is applied it will directly come to the output along with the clock.

1. Symbol



2. Truth Table

D	Q_{n+1}
0	0
1	1



Topic : D Flip Flop

3. Characteristics Table

$$Q_{n+1}(S, R, Q_n)$$
$$Q_{n+1}(J, K, Q_n)$$
$$Q_{n+1}(D, Q_n)$$

D	Q_n	Q_{n+1}
0	0	0
0	1	0
1	0	1
1	1	1

0
1
2
3

4. Characteristics Equation

$$Q_{n+1} = D$$

$$a = b$$
$$b = a$$

$$Q_{n+1} = \sum m(2, 3)$$
$$= D \cdot \bar{Q}_n + D \cdot Q_n = \underline{D}$$



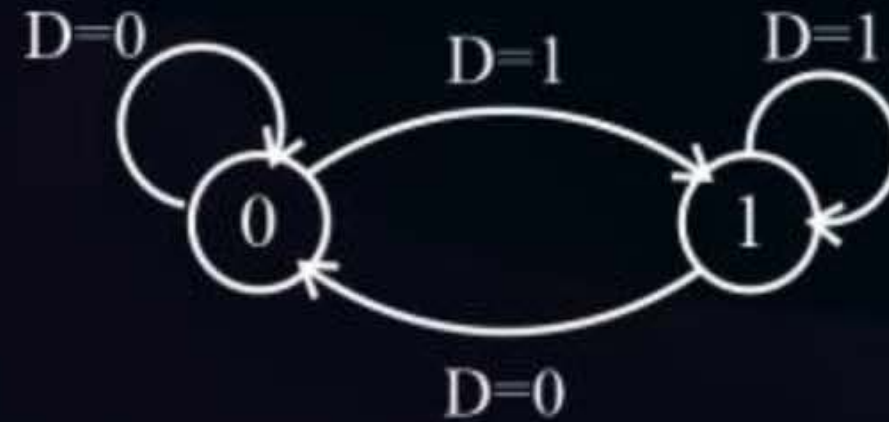
Topic : D Flip Flop

5. Excitation Table

$$\underline{D = Q_{n+1}}$$

Q_n	Q_{n+1}	D
0	0	0
0	1	1
1	0	0
1	1	1

6. State diagram





Topic : Sequential Circuit

$$Q_{n+1} = D$$

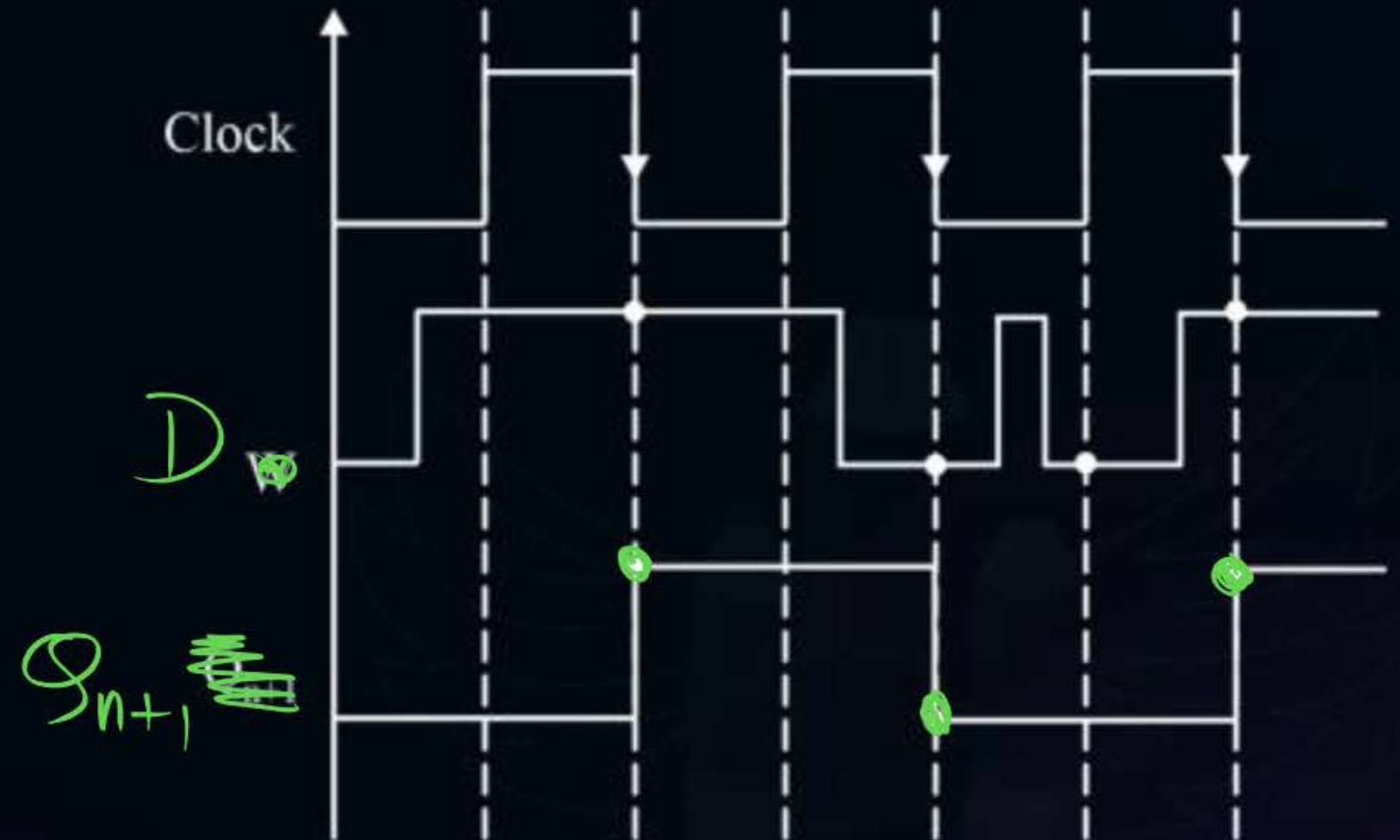




Topic : Sequential Circuit



edge triggered





Topic : Sequential Circuit

T FLIP FLOP (TOGGLE FLIP-FLOP)

1. Symbol





Topic : Sequential Circuit

T FLIP FLOP (TOGGLE FLIP-FLOP)

2. Truth table

T	Q_{n+1}
0	Q_n
1	$\bar{Q}_n$

→ Hold
→ Toggle



Topic : Sequential Circuit

T FLIP FLOP (TOGGLE FLIP-FLOP)

3. Characteristics Table

$$Q_{n+1} (T, Q_n)$$

T	Q_n	Q_{n+1}
0	0	0
0	1	1
1	0	1
1	1	0

} Hold $Q_{n+1} = Q_n$

} Toggle $Q_{n+1} = \overline{Q_n}$



Topic : Sequential Circuit

T FLIP FLOP (TOGGLE FLIP-FLOP)

4. Characteristics Equation

~~$$Q_{n+1} = \bar{T}Q_n + T\bar{Q}_n$$~~

$$Q_{n+1} = T \oplus Q_n$$

$$Q_{n+1} = \bar{T}Q_n + T\bar{Q}_n$$



Topic : Sequential Circuit

T FLIP FLOP (TOGGLE FLIP-FLOP)

5. Excitation table

Q_n	Q_{n+1}	T
0	0	0
0	1	1
1	0	1
1	1	0

$$Q_{n+1} = T \oplus Q_n$$

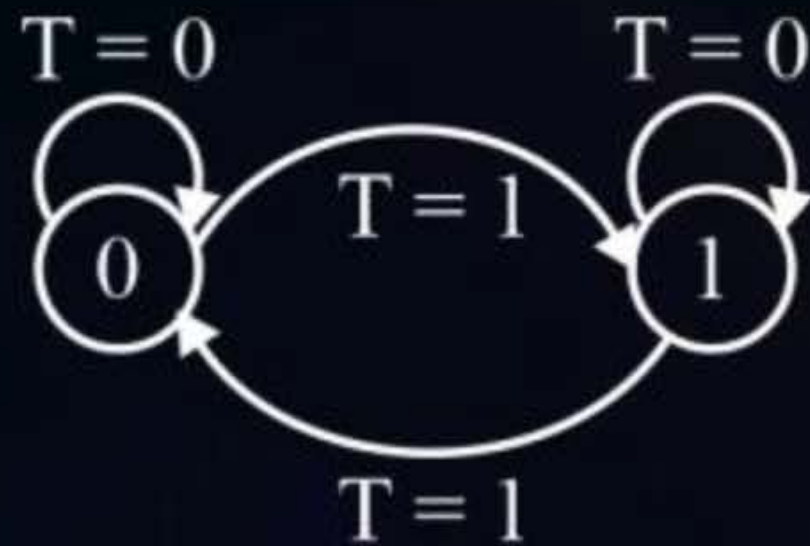
$$T = Q_{n+1} \oplus Q_n$$



Topic : Sequential Circuit

T FLIP FLOP (TOGGLE FLIP-FLOP)

6. State diagram





Topic : Sequential Circuit

Note: J-K FF is also known as universal FF.

SR FF

J	K	Q_{n+1}
0	0	Q_n
0	1	0
1	0	1
1	1	$\bar{Q}_n$

D FF

T FF



Topic : Sequential Circuit

SR FF

$$Q_{n+1} = S + \bar{R} Q_n$$

S	R	Q_{n+1}
0	0	Q_n
0	1	0
1	0	1
1	1	x

JK FF

$$Q_{n+1} = J \bar{Q}_n + \bar{K} Q_n$$

J	K	Q_{n+1}
0	0	Q_n
0	1	0
1	0	1
1	1	$\bar{Q}_n$

D FF

~~WFF~~

$$Q_{n+1} = D$$

D	Q_{n+1}
0	0
1	1

T - FF

$$Q_{n+1} = T \oplus Q_n$$

T	Q_{n+1}
0	Q_n
1	$\bar{Q}_n$



Topic : Sequential Circuit

Excitation Table



0
1
2
3

Q_n	Q_{n+1}	0 S	R	J	K	0 D	T
0	0	0	X	0	X	0	0
0	1	1	0	1	X	1	1
1	0	0	1	X	1	0	1
1	1	X	0	X	0	1	0



Topic : Sequential Circuit

Designing of Flip-Flop:

Step 1: → Write the characteristic Table of desired FF. →

Step 2: → Write the excitation table of available FF. →

CD
EA

Step 3: → Write the logical expression.

Step 4: → Minimize the logical expression.

Step 5: → Hardware Implementation.

Available (given) → Desired (Reqd)

① Design JK using SR

CDCA
↓

Desired $\rightarrow$ JK $\rightarrow$ Characteristic table

Available $\rightarrow$ SR $\rightarrow$ Excitation table.



Topic : Sequential Circuit

Design JK FF by using SR FF?

Step-1 $\rightarrow$ Characteristic table of JK

Step-2

Excitation table of SR

$S \rightarrow (J, K, Q_n)$
 $R \rightarrow (J, K, Q_n)$

$Q_{n+1}(J, K, Q_n)$

Step-1

Step-2

$S, R \Rightarrow F(Q_n, Q_{n+1})$

	J	K	Q_n	Q_{n+1}	S	R
0	0	0	0	0	0	X
1	0	0	1	1	X	0
2	0	1	0	0	0	X
3	0	1	1	0	0	1
4	1	0	0	1	1	0
5	1	0	1	1	X	0
6	1	1	0	1	1	0
7	1	1	1	0	0	1

Excitation $\rightarrow$ SR

Q_n	Q_{n+1}	S	R
0	0	0	X
0	1	1	0
1	0	0	1
1	1	X	0



Topic : Sequential Circuit

Step-3

$$S(J, K, Q_n) = \sum (4, 6) + \sum d (1, 5)$$

$$R(J, K, Q_n) = \sum m (3, 7) + \sum d (0, 2)$$

JK using SR

Step-4 (minimize)

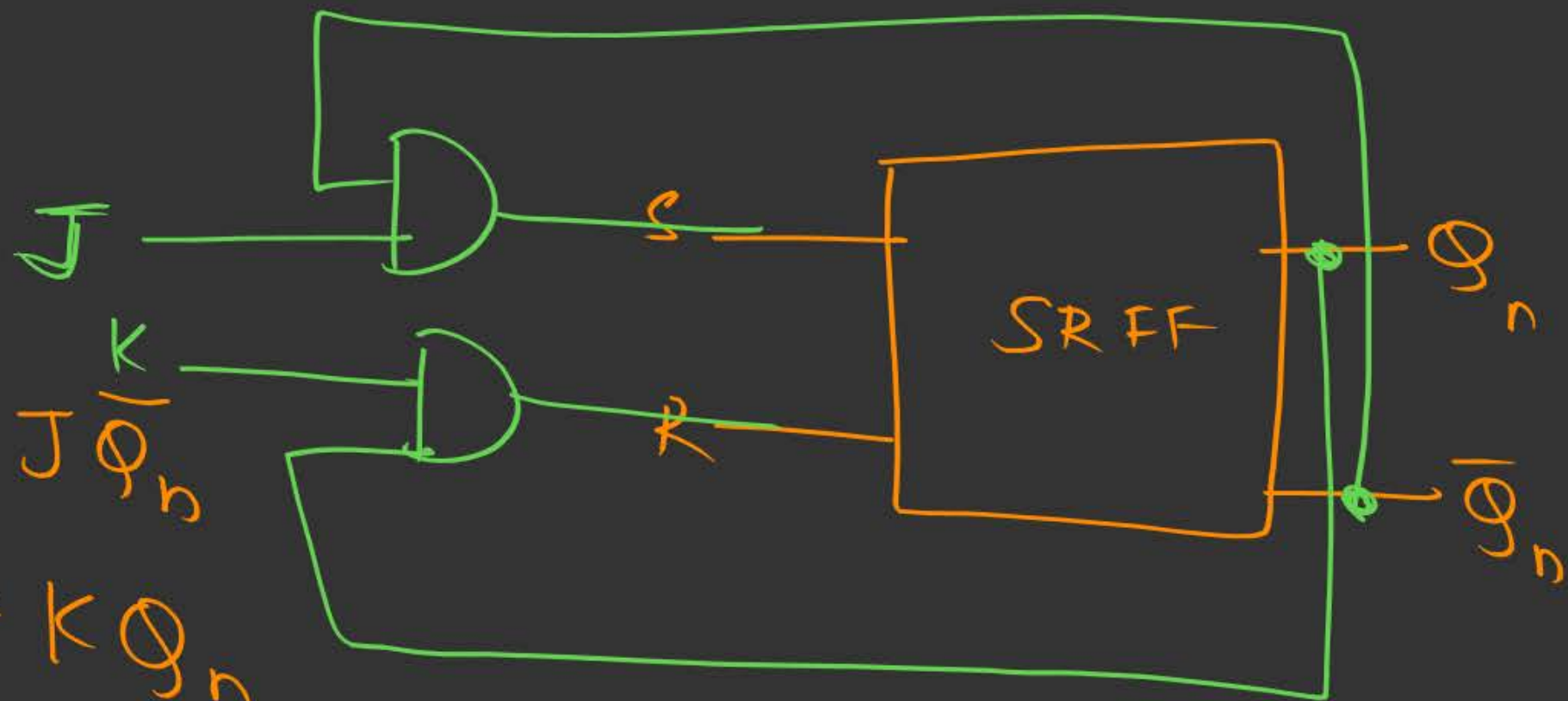
		Q_n			
		$\bar{1}0$	$\bar{1}1$	11	10
J	0	0	x	3	2
	1	1	x	7	6

$$S = J \bar{Q}_n$$

		Q_n			
		$\bar{1}0$	$\bar{1}1$	11	10
K	0	x	1	1	x
	1	4	5	1	6

$$R = K Q_n$$

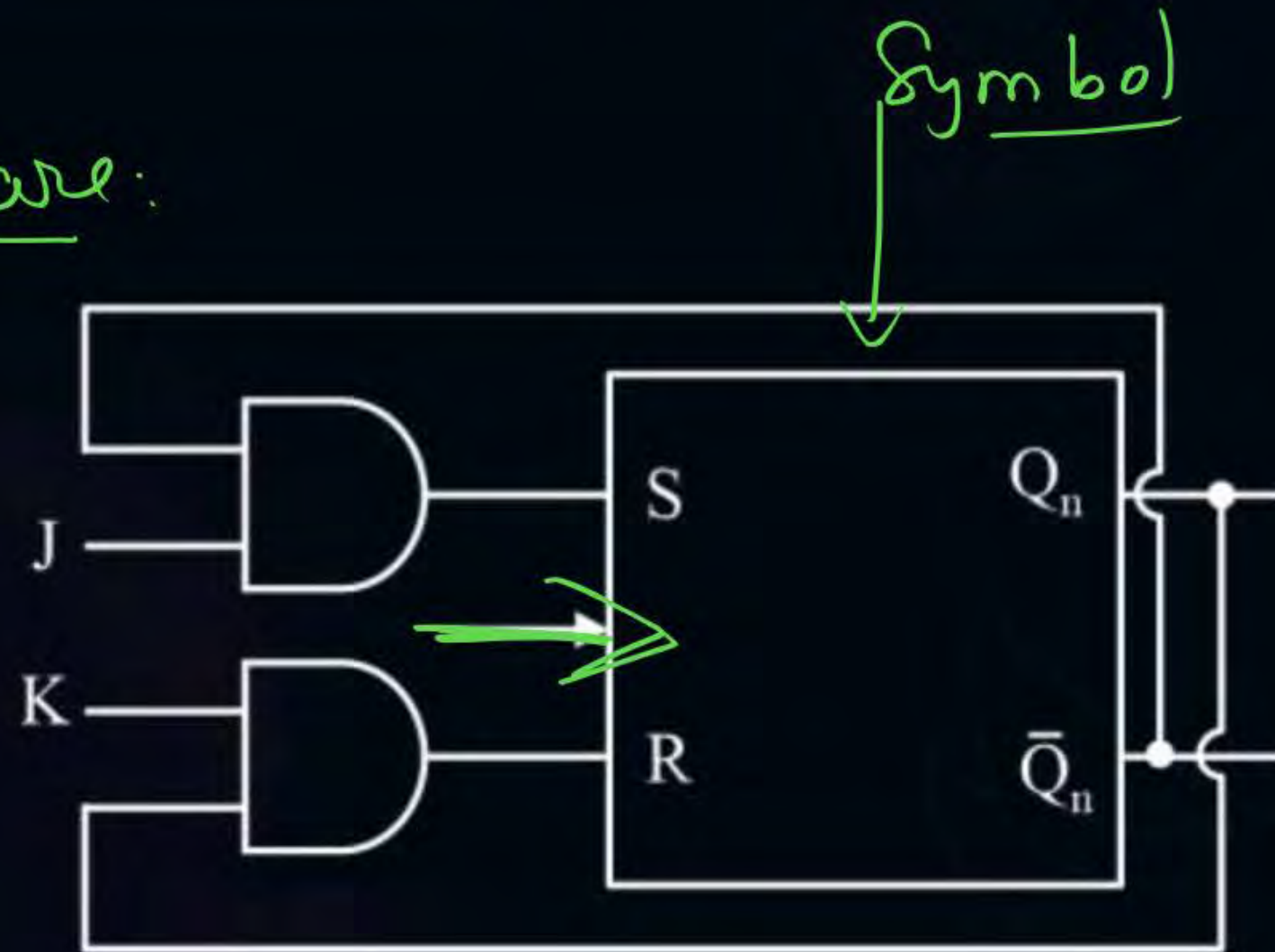
$$S = J \overline{Q_n}$$
$$R = K Q_n$$





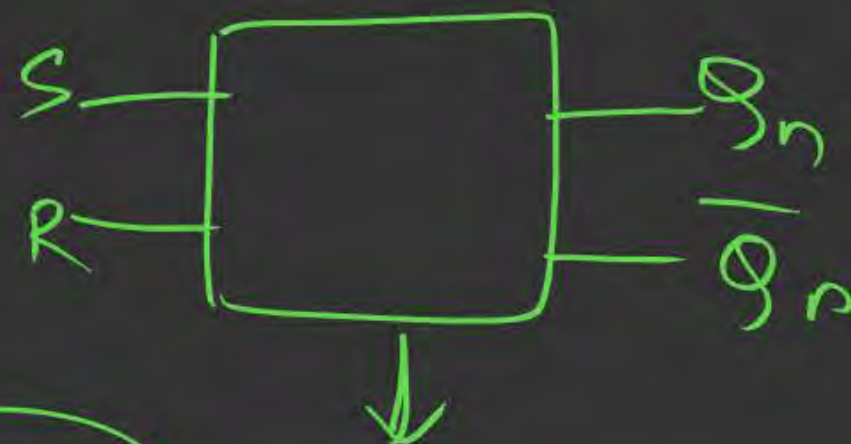
Topic : Sequential Circuit

Step-5 : Hardware:

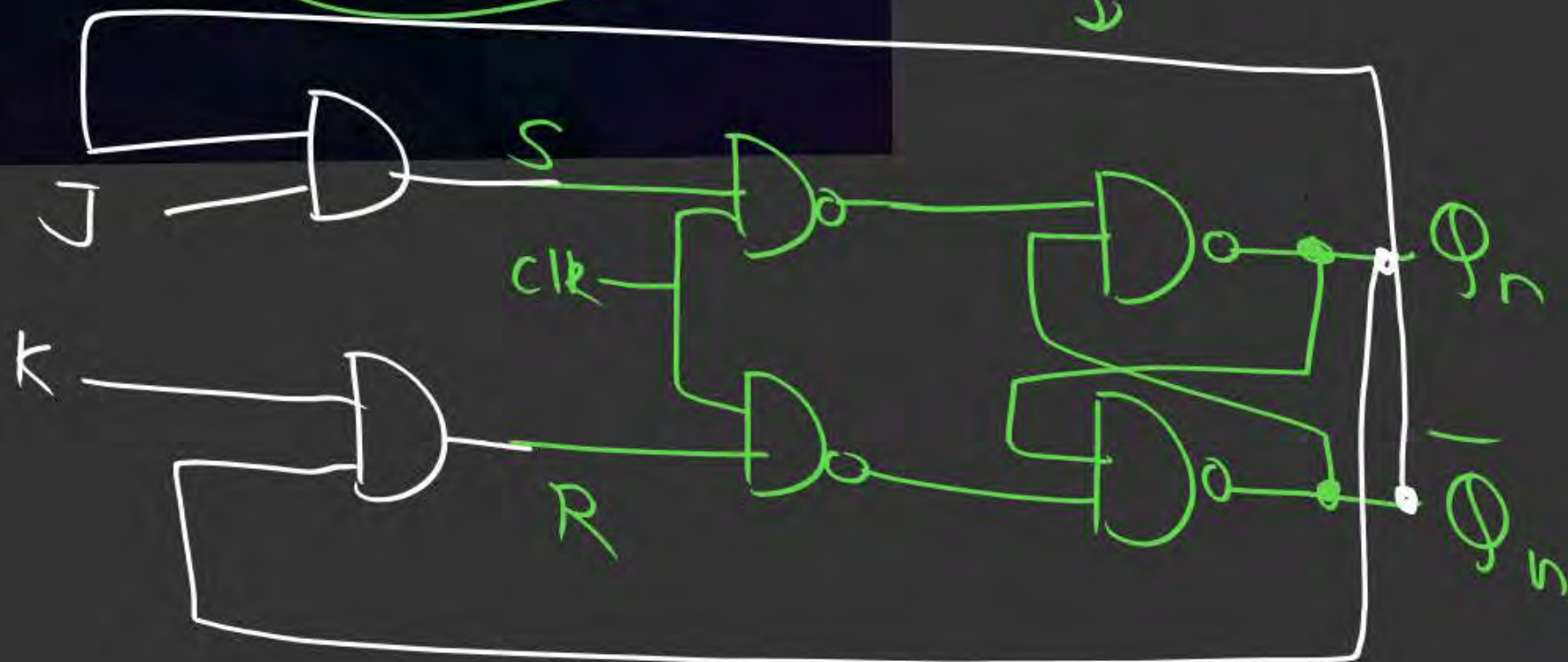




Step-5



Circuit Diagram
of
JK using
SR

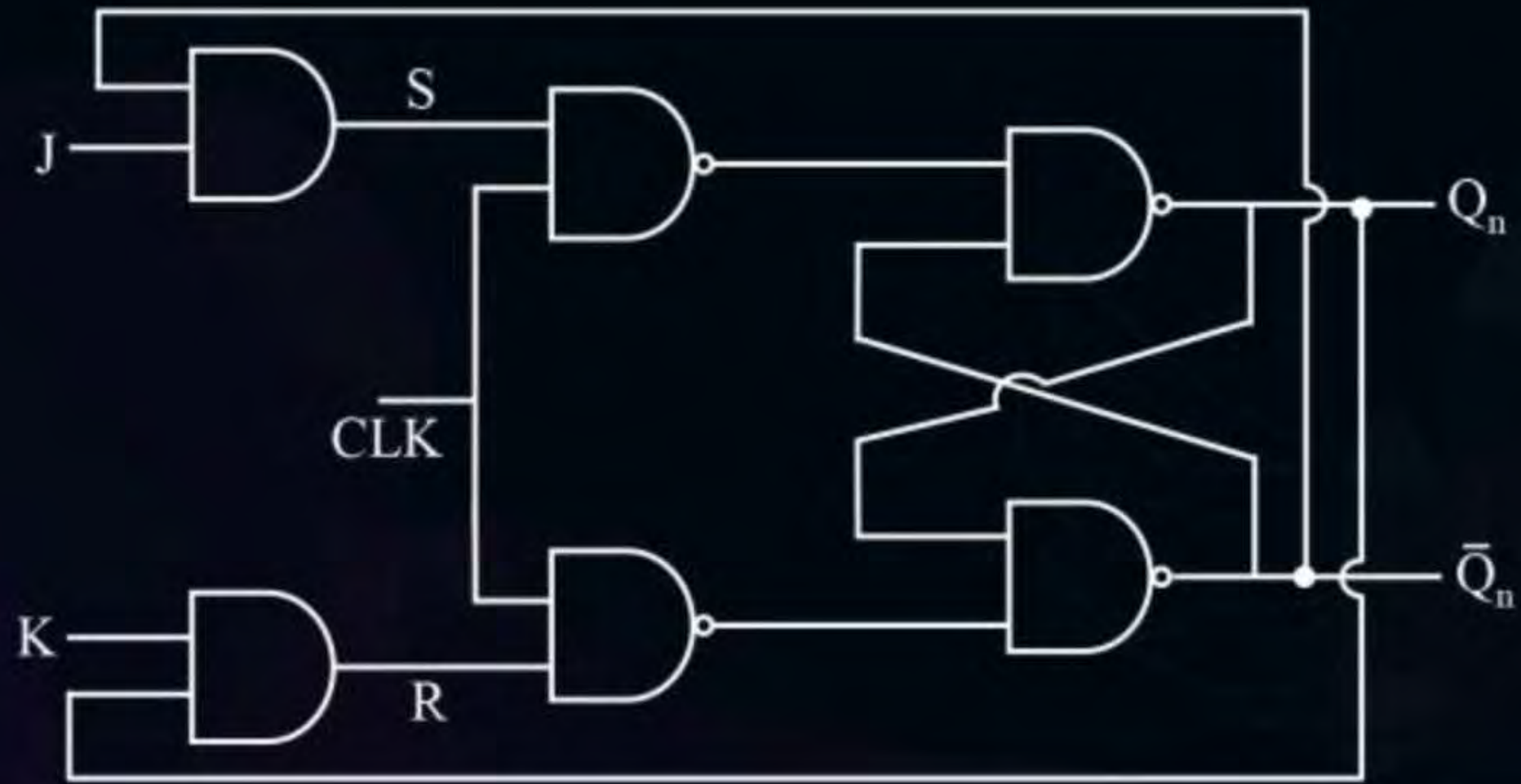




Topic : Sequential Circuit



0,0
😊



(Q) Design a D FF using SR FF

Desired $\rightarrow$ D FF $\rightarrow$ Characteris table

Available $\rightarrow$ SR FF $\rightarrow$ Excitation table

#Q. Design a w-FF by using SR FF ?

Step 1 $\rightarrow Q_{n+1}(D, Q_n)$

Step 2

$\downarrow$
S, R Excitation

$S, R \rightarrow F(Q_n, Q_{n+1})$

0
1
2
3

Step 1			Step 2	
D	Q_n	Q_{n+1}	S	R
0	0	0	0	X
0	1	0	0	1
1	0	1	1	0
1	1	1	X	0

Q_n	Q_{n+1}	S	R
0	0	0	X
0	1	1	0
1	0	0	1
1	1	X	0

Step 3

$$S = \sum m(2) + \sum d(3)$$

$$R = \sum m(1) + \sum d(0)$$

$S \Rightarrow \overline{D}$
 D

Q_n	$\overline{Q}_n$	Q_n
$\overline{Q}_n$	0	1
0	0	1
1	1	X
	2	3

$$\underline{\underline{S = D}}$$

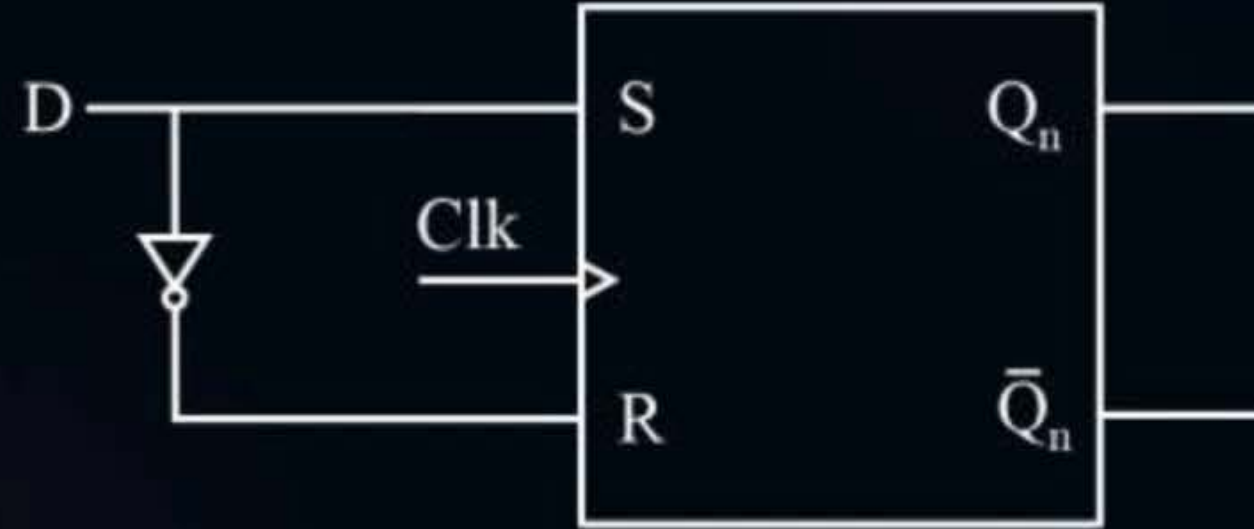
Step 4

$R \Rightarrow \overline{D}$
 D

Q_n	$\overline{Q}_n$	Q_n
$\overline{Q}_n$	0	1
0	X	1
1		
	2	3

$$R = \overline{D}$$

Step 5 Handwore



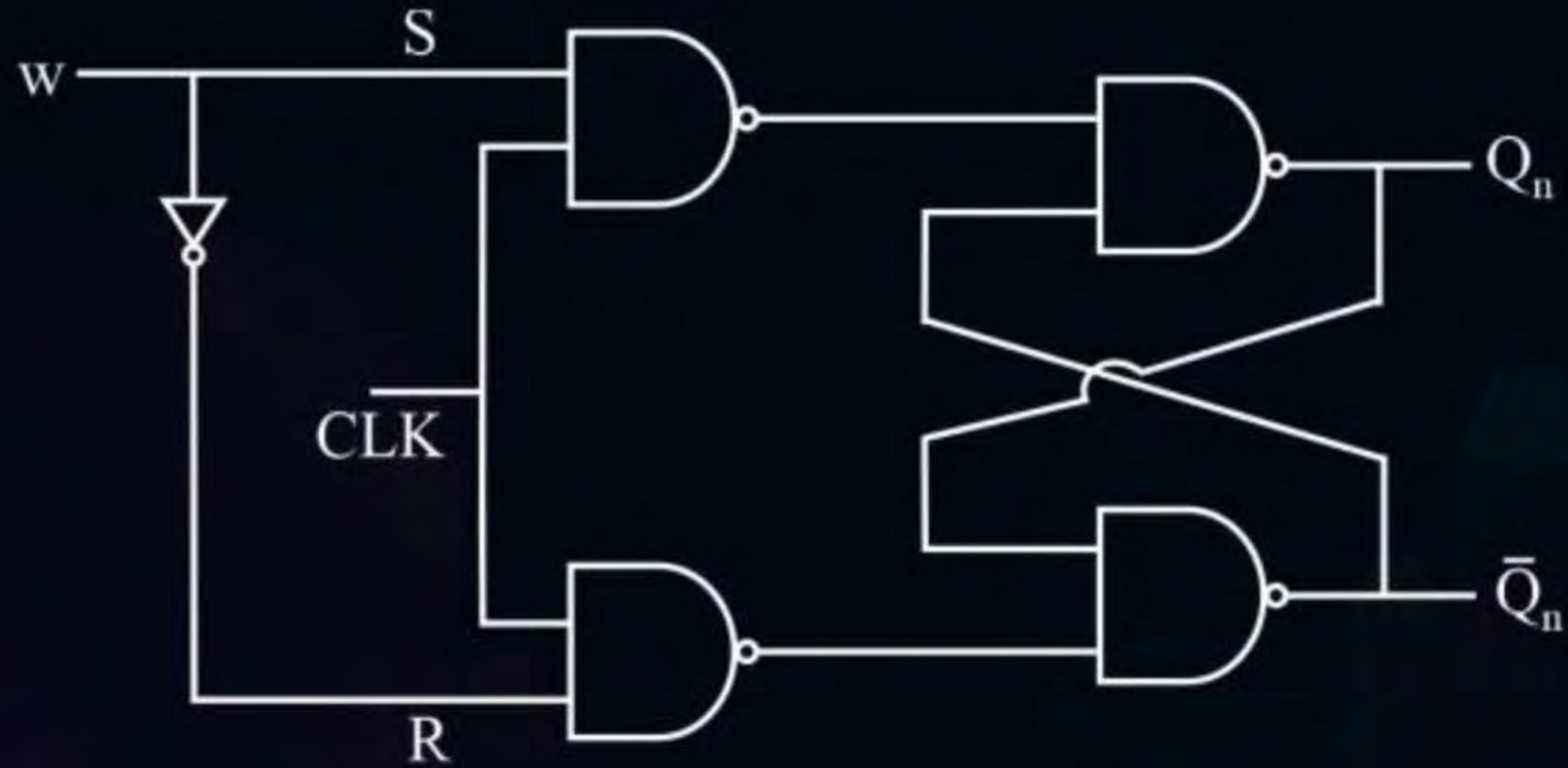
$$S = D$$

$$R = \overline{D}$$

H.W: Design Circuit



Topic : Sequential Circuit





2 mins Summary



- ✓ Master Slave
- ✓ D FF
- ✓ T FF
- ✓ Design Framework for FF



THANK - YOU